

semmcci: Staging

Ivan Jacob Agaloos Pesigan

```
library(semmcci)
library(lavaan)

# Data -----
data("Tal.Or", package = "psych")
df <- mice::ampute(Tal.Or)$amp

# Monte Carlo -----
## Fit Model in lavaan -----
model <- "
  reaction ~ cp * cond + b * pmi
  pmi ~ a * cond
  cond ~~ cond
  indirect := a * b
  direct := cp
  total := cp + (a * b)
"
fit <- sem(data = df, model = model, missing = "fiml")

## MC() -----
unstd <- MC(
  fit,
  R = 100L, # use a large value e.g., 20000L for actual research
  alpha = 0.05
)

## Standardized Monte Carlo -----
MCStd(unstd, alpha = 0.05)

#> Standardized Monte Carlo Confidence Intervals
#>
#>      est      se    R    2.5%  97.5%
#> cp      0.1007 0.0930 100 -0.0907 0.2452
#> b      0.4437 0.0845 100  0.2525 0.5897
#> a      0.1730 0.0903 100 -0.0183 0.3532
#> cond~~cond 1.0000 0.0000 100  1.0000 1.0000
#> reaction~~reaction 0.7775 0.0712 100  0.6300 0.8839
```

```

#> pmi~~pmi          0.9701 0.0315 100  0.8753 0.9996
#> indirect          0.2243 0.0412 100 -0.0085 0.1513
#> direct            4.1748 0.0930 100 -0.0907 0.2452
#> total             0.9194 0.0997 100 -0.0404 0.3487

# Monte Carlo (Multiple Imputation) -----
## Multiple Imputation -----
mi <- mice::mice(
  data = df,
  print = FALSE,
  m = 5L, # use a large value e.g., 100L for actual research,
  seed = 42
)

## Fit Model in lavaan -----
fit <- sem(data = df, model = model) # use default listwise deletion

## MCMI() -----
unstd <- MCMI(
  fit,
  mi = mi,
  R = 100L, # use a large value e.g., 20000L for actual research
  alpha = 0.05
)

## Standardized Monte Carlo -----
MCStd(unstd, alpha = 0.05)

#> Standardized Monte Carlo Confidence Intervals
#>
#>          est      se    R    2.5%  97.5%
#> cp          0.0885 0.0862 100 -0.0322 0.2702
#> b           0.4214 0.0843 100  0.2399 0.5693
#> a           0.1412 0.0879 100  0.0361 0.3440
#> cond~~cond   1.0000 0.0000 100  1.0000 1.0000
#> reaction~~reaction 0.8040 0.0735 100  0.6219 0.8997
#> pmi~~pmi     0.9801 0.0329 100  0.8814 0.9979
#> indirect     0.0595 0.0441 100  0.0101 0.1698
#> direct       0.0885 0.0862 100 -0.0322 0.2702
#> total        0.1480 0.0859 100  0.0275 0.3474

```

References

- Pesigan, I. J. A., & Cheung, S. F. (2024). Monte Carlo confidence intervals for the indirect effect with missing data. *Behavior Research Methods*, 56(3), 1678–1696. <https://doi.org/10.3758/s13428-023-02114-4>
- R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>